

Evaluation of 6 Models for BigCodeBench/704

Task Overview

Problem Description: Find all combinations of columns from a given DataFrame so that the absolute correlation between them is greater than a certain threshold. The function should output a list of tuples where each tuple contains two column names.

Model 1 Evaluation

model1_efficiency

Total steps: 6

Step Analysis:

- <step 1> Problem understanding - **Essential**
- <step 2> Correlation matrix approach - **Essential**
- <step 3> Using itertools.combinations - **Essential**
- <step 4> Accessing correlation values - **Essential**
- <step 5> Absolute value and threshold check - **Essential**
- <step 6> Building and returning result - **Essential**

Efficiency Score: 6/6 essential steps. Well-structured and focused.

model1_logic

Step Coherence:

- All steps flow logically
- Clear progression: problem → correlation matrix → combinations → filtering → result
- Each step builds on the previous one

Internal Consistency:

- Consistent approach throughout
- Correctly identifies need for absolute values
- No contradictions

model1_completeness

Problem Coverage:

Problem Component	Addressed?
Find column combinations	Yes
Calculate correlations	Yes
Use absolute correlation	Yes
Apply threshold (>)	Yes
Return list of tuples	Yes
Use specified columns only	Yes

Factual Correctness:

- Correct understanding of correlation matrix
- Accurate use of itertools.combinations
- Proper handling of symmetric correlation

Edge Case Handling:

- No explicit handling of NaN values
- No validation of input columns
- No consideration of empty results

Model 2 Evaluation

model2_efficiency

Total steps: 6

Step Analysis:

- <step 1> Requirements analysis - **Essential**
- <step 2> Using itertools.combinations - **Essential**
- <step 3> Correlation calculation method - **Essential**
- <step 4> Absolute value and threshold check - **Essential**
- <step 5> Adding to results - **Non-essential** (redundant with step 6)
- <step 6> Return statement - **Essential**

Efficiency Score: 5/6 essential steps. Step 5 is redundant.

model2_logic

Step Coherence:

- Logical flow throughout

- Uses Series.corr() method instead of correlation matrix
- Steps 5 and 6 overlap in content

Internal Consistency:

- Consistent approach
- Different method (pairwise correlation) but valid
- No contradictions

model2_completeness

Problem Coverage:

Problem Component	Addressed?
Find column combinations	Yes
Calculate correlations	Yes
Use absolute correlation	Yes
Apply threshold (>)	Yes
Return list of tuples	Yes
Use specified columns only	Yes

Factual Correctness:

- Correct use of Series.corr() method
- Accurate understanding of combinations
- Proper absolute value handling

Edge Case Handling:

- No NaN handling mentioned
- No input validation
- No error handling

Model 3 Evaluation

model3_efficiency

Total steps: 8

Step Analysis:

- <step 1> Problem statement - **Essential**
- <step 2> Correlation matrix approach - **Essential**

- <step 3> Detailed combinations explanation - **Essential** (though verbose)
- <step 4> Iteration through pairs - **Essential**
- <step 5> Absolute value requirement - **Essential**
- <step 6> Threshold comparison - **Non-essential** (could be combined with step 5)
- <step 7> Result collection - **Essential**
- <step 8> Final return and edge cases - **Essential**

Efficiency Score: 7/8 essential steps. Most detailed explanation.

model3_logic

Step Coherence:

- Excellent logical flow
- Very detailed explanation of why combinations is appropriate
- Step 6 could be merged with step 5

Internal Consistency:

- Highly consistent
- Thorough explanation of symmetry
- Mentions NaN handling at the end

model3_completeness

Problem Coverage:

Problem Component	Addressed?
Find column combinations	Yes
Calculate correlations	Yes
Use absolute correlation	Yes
Apply threshold (>)	Yes
Return list of tuples	Yes
Use specified columns only	Yes

Factual Correctness:

- Excellent technical accuracy
- Correct explanation of correlation symmetry
- Proper understanding of combinations

Edge Case Handling:

- **Best edge case consideration:** Mentions NaN handling
 - Notes that NaN comparisons will be false
 - Most comprehensive edge case awareness
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Model 4 Evaluation

model4_efficiency

Total steps: 7

Step Analysis:

- <step 1> Threshold calculation with MIN_PERCENTAGE - **Essential** (but introduces confusion)
- <step 2> DataFrame subsetting - **Essential**
- <step 3> Correlation matrix - **Essential**
- <step 4> Upper triangle iteration - **Essential**
- <step 5> NaN check and threshold - **Essential**
- <step 6> Result collection - **Essential**
- <step 7> Return statement - **Essential**

Efficiency Score: 7/7 essential steps. Includes implementation.

model4_logic

Step Coherence:

- Generally coherent but step 1 introduces unexpected logic (percentage * MIN_PERCENTAGE)
- Upper triangle approach is valid alternative to combinations
- Implementation matches description

Internal Consistency:

- **Major inconsistency:** Uses $\text{percentage} * \text{MIN_PERCENTAGE}$ which differs from problem description
- Otherwise consistent approach
- Code implementation aligns with steps

model4_completeness

Problem Coverage:

Problem Component	Addressed?
Find column combinations	Yes
Calculate correlations	Yes
Use absolute correlation	Yes
Apply threshold	Partial (modified threshold)
Return list of tuples	Yes
Use specified columns only	Yes

Factual Correctness:

- Correct correlation matrix usage
- Valid upper triangle iteration
- **Error:** Unexplained threshold modification

Edge Case Handling:

- Explicit NaN checking with `pd.notnull()`
- Good implementation of edge case handling
- But threshold modification is problematic

Model 5 Evaluation

model5_efficiency

Total steps: 4

Step Analysis:

- <step 1> Correlation matrix computation - **Essential**
- <step 2> Using combinations for unique pairs - **Essential**
- <step 3> Threshold comparison and result collection - **Essential**
- <step 4> Edge case discussion - **Essential**

Efficiency Score: 4/4 essential steps. Most concise approach.

model5_logic

Step Coherence:

- Clear logical progression
- Concise but complete
- All steps necessary and coherent

Internal Consistency:

- Fully consistent
- No contradictions
- Clear about "greater than" vs "greater than or equal to"

model5_completeness

Problem Coverage:

Problem Component	Addressed?
Find column combinations	Yes
Calculate correlations	Yes
Use absolute correlation	Yes
Apply threshold (>)	Yes
Return list of tuples	Yes
Use specified columns only	Yes

Factual Correctness:

- All statements accurate
- Correct understanding of requirements
- Proper threshold interpretation

Edge Case Handling:

- Mentions NaN handling
- Notes fewer than two columns case
- Clarifies strict inequality requirement

Model 6 Evaluation

model6_efficiency

Total steps: 6

Step Analysis:

- <step 1> Problem understanding - **Essential**
- <step 2> Input/output identification - **Essential**
- <step 3> Combinations approach - **Essential**
- <step 4> Correlation calculation - **Essential**

- <step 5> Absolute value and threshold check - **Essential**
- <step 6> Return results - **Essential**

Efficiency Score: 6/6 essential steps. Standard approach.

model6_logic

Step Coherence:

- Clear logical flow
- Each step builds appropriately
- Standard progression through solution

Internal Consistency:

- Consistent throughout
- No contradictions
- Clear explanation of absolute value reasoning

model6_completeness

Problem Coverage:

Problem Component	Addressed?
Find column combinations	Yes
Calculate correlations	Yes
Use absolute correlation	Yes
Apply threshold (>)	Yes
Return list of tuples	Yes
Use specified columns only	Yes

Factual Correctness:

- Correct understanding of all components
- Accurate description of Pearson correlation
- Proper use of combinations

Edge Case Handling:

- No edge cases considered
 - No NaN handling mentioned
 - No input validation
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Summary Ranking

Best Overall: Model 3 - Most thorough with edge case consideration **Most Efficient:** Model 5 - Only 4 essential steps while covering all aspects **Most Problematic:** Model 4 - Introduces unexplained threshold modification **Best Edge Case Handling:** Models 3 and 5 - Both consider NaN values **Clearest Logic:** Models 1, 3, and 5 - All have excellent logical flow

Key Observations:

- Most models use correlation matrix approach; Model 2 uses pairwise correlation
- Model 4 introduces confusing threshold calculation (percentage * 0.75)
- Models 3 and 5 are the only ones to explicitly consider edge cases
- All models correctly understand the need for absolute correlation values
- Model 4 uses upper triangle iteration instead of combinations (valid alternative)